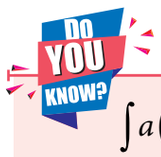
Example 2.2

Evaluate $\int \sqrt{2x+1}\,dx$

Solution:

$$\int \sqrt{2x+1}\,dx = \int (2x+1)^{\frac{1}{2}}\,dx$$

$$= \frac{(2x+1)^{\frac{3}{2}}}{3} + c$$



$$\int a(ax+b)^n\,dx = \int (ax+b)^n\,d(ax+b)$$

$$= \frac{(ax+b)^{n+1}}{n+1} + c$$

Example 2.3

Evaluate $\int \frac{dx}{(2x+3)^2}$

Solution:

$$\int \frac{dx}{(2x+3)^2} = \int (2x+3)^{-2}\,dx$$

$$= -\frac{1}{2(2x+3)} + c$$



(i) $\int \frac{1}{x^n}\,dx = -\frac{1}{(n-1)x^{n-1}} + c\ ,\ n \neq 1$

(ii) $\int \frac{a}{(ax+b)^n}\,dx = \int \frac{d(ax+b)}{(ax+b)^n}$

$$= -\frac{1}{(n-1)(ax+b)^{n-1}} + c\ ,\ n \neq 1$$

Example 2.4

Evaluate $\int \left(x + \frac{1}{x}\right)^2\,dx$

Solution:

$$\int \left(x + \frac{1}{x}\right)^2\,dx = \int \left(x^2 + 2 + \frac{1}{x^2}\right)dx$$

$$= \int x^2\,dx + 2\int dx + \int \frac{1}{x^2}\,dx$$

$$= \frac{x^3}{3} + 2x - \frac{1}{x} + c$$

Example 2.5

Evaluate $\int (x^3+7)(x-4)\,dx$

Solution:

$$\int (x^3+7)(x-4)\,dx$$

$$= \int (x^4 - 4x^3 + 7x - 28)\,dx$$

$$= \frac{x^5}{5} - x^4 + \frac{7x^2}{2} - 28x + c$$

Example 2.6

Evaluate $\int \frac{2x^2 - 14x + 24}{x-3}\,dx$

Solution:

By factorisation,

$$2x^2 - 14x + 24 = (x-3)(2x-8)$$

$$\int \frac{2x^2 - 14x + 24}{x-3}\,dx = \int \frac{(x-3)(2x-8)}{x-3}\,dx$$

$$= \int (2x-8)\,dx$$

$$= x^2 - 8x + c$$

Example 2.7

Evaluate $\int \frac{x+2}{\sqrt{2x+3}}\,dx$

Solution:

Split into simple integrands

$$\frac{x+2}{\sqrt{2x+3}} = \frac{\frac{1}{2}(2x+4)}{(2x+3)^{\frac{1}{2}}}$$

$$= \frac{1}{2}\left\{\frac{(2x+3)+1}{(2x+3)^{\frac{1}{2}}}\right\}$$

$$= \frac{1}{2}\left\{(2x+3)^{\frac{1}{2}} + (2x+3)^{-\frac{1}{2}}\right\}$$

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